

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 1

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Score:	_____	Date:	_____
		Instructor's Signature:	_____

Lab # 1: To Configure the Security Policies on Firewall

Objective: The objective of this lab is to configure Security Policies on the Firewall. **Duration: 3 Hours**

Expected Outcomes: By the end of this lab session, students will be able to:

1. Understand the principles of Security Policies
2. Understanding the relationship between different security policies
3. Configure firewall security policies

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Topology

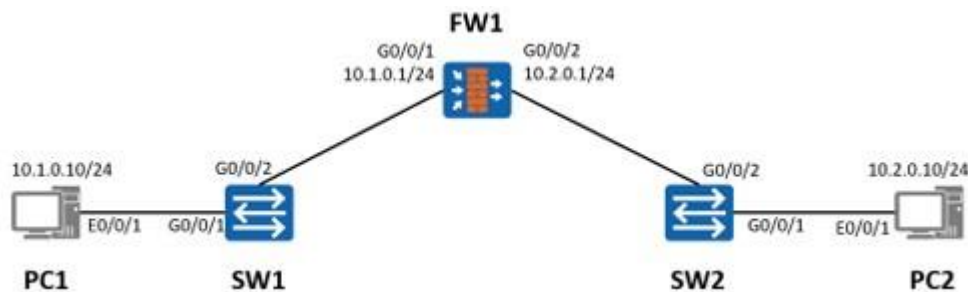


Figure 1 Configuring Firewall Security Policies

Lab Planning

Device	Interface	IP Address	Security Zone
FW1	GigabitEthernet 1/0/0	10.1.0.1/24	Trust
	GigabitEthernet 1/0/2	10.2.0.1/24	Untrust
PC1	Eth 0/0/1	10.1.0.10/24	Trust

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PC2	Eth 0/0/1	10.2.0.10/24	Untrust
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Configuration Roadmap

1. Configure basic IP addresses and security zones.
2. Configure an interzone security policy.
3. Configure the gateways of PC1 and PC2 to use IP addresses that are on the same network segment as those of the corresponding interfaces on the firewall.

Configuration Procedure on the CLI

Step 1 Complete the configurations of the upstream and downstream interfaces of FW1.

Configure the IP addresses for interfaces and add them to security zones.

```
[FW1] interface G1/0/0
[FW1-GigabitEthernet1/0/0] ip address 10.1.0.1 255.255.255.0
[FW1-GigabitEthernet1/0/0] quit
[FW1] interface G1/0/2
[FW1-GigabitEthernet1/0/2] ip address 10.2.0.1 255.255.255.0
[FW1-GigabitEthernet1/0/2] quit
[FW1] firewall zone trust
[FW1-zone-trust] add interface G1/0/0
[FW1-zone-trust] quit
[FW1] firewall zone untrust
[FW1-zone-untrust] add interface G1/0/2
[FW1-zone-untrust] quit
```

Step 2 Configure a forwarding policy between the Trust zone and the Untrust zone.

```
[FW1] security-policy
[FW1-policy-security] rule name policy_sec
[FW1-policy-security-rule-policy_sec] source-zone trust
[FW1-policy-security-rule-policy_sec] destination-zone untrust
[FW1-policy-security-rule-policy_sec] action permit
[FW1-policy-security-rule-policy_sec] quit
```

Step 3 Configure the PC.

Set the IP address of PC1 to 10.1.0.10/24 and that of the gateway to 10.1.0.1. Set the IP address of PC2 to 10.2.0.10/24 and that of the gateway to 10.2.0.1.

Verification

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Ping **10.2.0.10** on the CLI from PC1 to check whether PC1 can ping PC2 (Attach screenshots)

```
PC> ping 10.2.0.10
Ping 10.2.0.10: 32 data bytes, Press Ctrl_C to break
From 10.2.0.10: bytes=32 seq=1 ttl=127 time=16 ms
```

Run the display firewall session table command to view the session table of the firewall.

```
[FW1] display firewall session table
Current Total Sessions : 1
icmp VPN: public --> public 10.1.0.10:49569 --> 10.2.0.10:2048
```

Lab Task:

1. Role of the “trust” zone and the “untrust” zone

- The **trust zone** represents the **internal, secure network**.
 - Devices in this zone (e.g., **PC1 – 10.1.0.10**) are considered reliable.
- The **untrust zone** represents the **external or less secure network**.
 - Devices in this zone (e.g., **PC2 – 10.2.0.10**) are considered untrusted.

The firewall uses these zones to **control traffic flow** based on security policies. Traffic between zones is **blocked by default** unless explicitly permitted.

2. Meaning of the command [FW1-policy-security] rule name policy_sec

This command **creates a new security policy rule** named **policy_sec** inside the firewall’s security policy configuration mode.

- It defines a **logical container** where conditions (source zone, destination zone) and actions (permit/deny) are specified.
- The rule name helps in **identification, management, and troubleshooting** of firewall policies.

3. Significance of source-zone trust and destination-zone untrust

- source-zone trust specifies that traffic **originates from the trusted network**.
- destination-zone untrust specifies that traffic is **going toward the untrusted network**.

Type of traffic permitted by this rule:

- Any traffic initiated from the **Trust zone to the Untrust zone** (e.g., ICMP ping, HTTP, etc.).
- In this lab, it allows **PC1 (10.1.0.10)** to successfully **ping PC2 (10.2.0.10)**.

This follows the **principle of least privilege**, allowing internal users to access external networks while keeping the internal network protected.

4. Allowing traffic from “untrust” zone to “trust” zone

To allow traffic from **Untrust → Trust**:

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- You should **create a new security policy rule**, not modify the existing one.
- Reason: Modifying the existing rule would change its intended direction and reduce clarity.

Example configuration:

[FW1] security-policy

[FW1-policy-security] rule name untrust_to_trust

[FW1-policy-security-rule-untrust_to_trust] source-zone untrust

[FW1-policy-security-rule-untrust_to_trust] destination-zone trust

[FW1-policy-security-rule-untrust_to_trust] action permit

This explicitly allows traffic initiated from the untrusted network to reach the trusted network.

5. Role of display firewall session table command

- This command shows **active sessions** currently passing through the firewall.
- It helps verify:
 - Whether traffic is being **permitted or blocked**
 - Whether security policies are **working correctly**
 - Source IP, destination IP, protocol, and session state

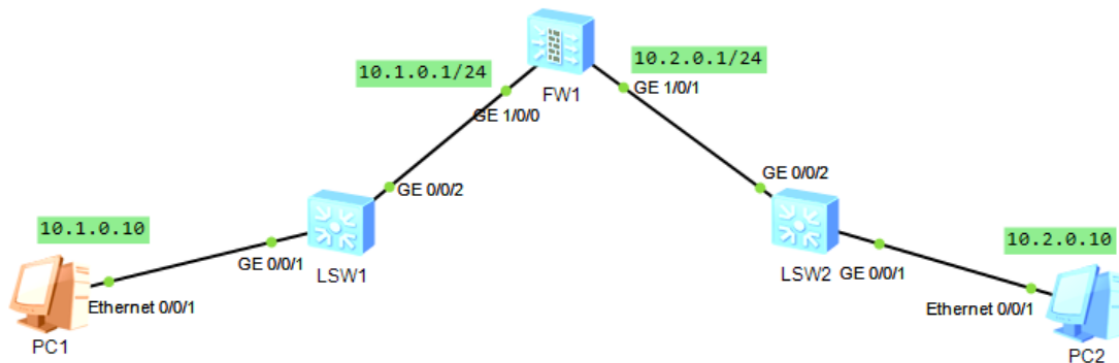
Meaning of the output:

icmp VPN: public --> public 10.1.0.10:49569 -> 10.2.0.10:2048

- **icmp**: The protocol used (ping)
- **10.1.0.10 → 10.2.0.10**: Traffic from PC1 to PC2
- This entry confirms that:
 - The firewall **allowed the ICMP session**
 - The configured security policy **policy_sec** is effective

Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-23BSCYS001

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```
[FW-23BSCYS006]display security-policy rule all
```

```
2025-12-29 00:20:11.590
```

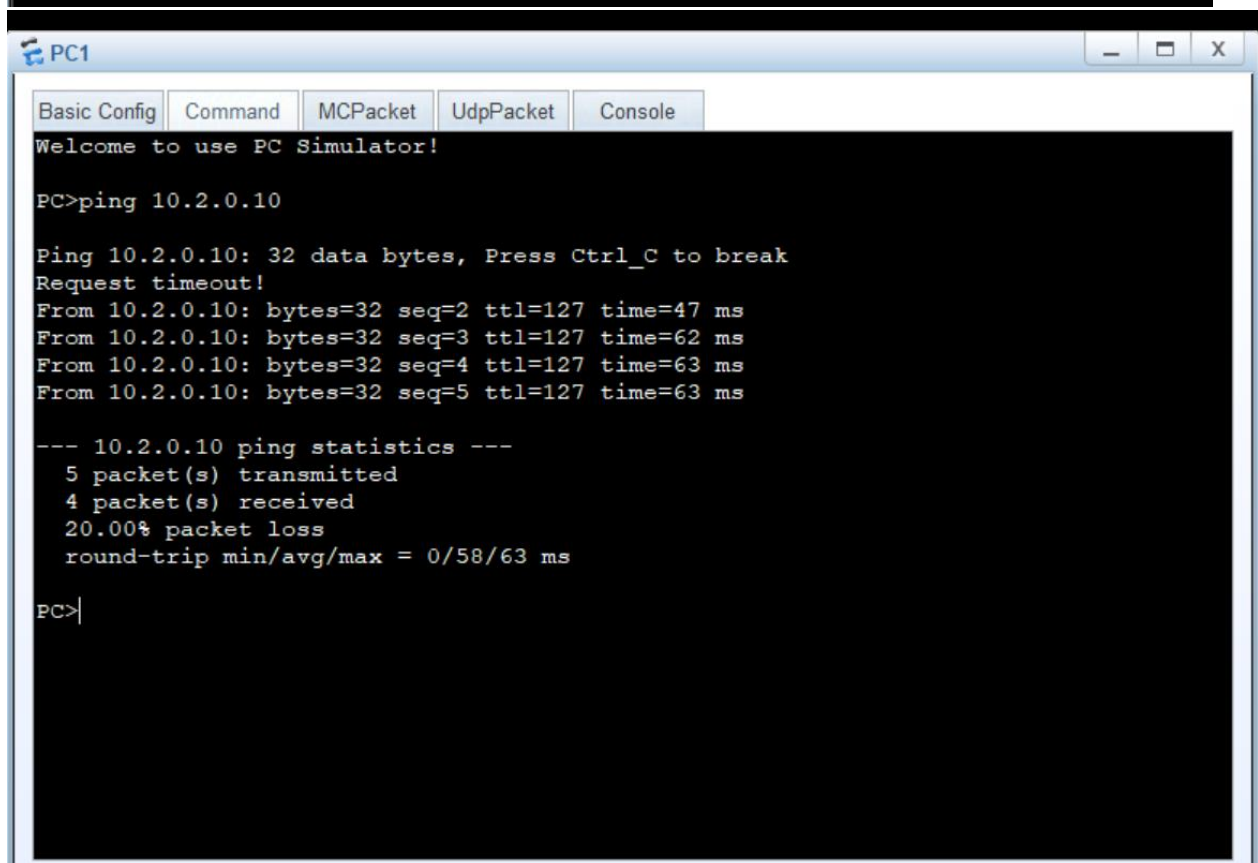
```
Total:2
```

RULE ID	RULE NAME	STATE	ACTION	HITS
1	policy_sec	enable	permit	5
0	default	enable	deny	0

```
[FW-23BSCYS006]
```

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Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	192.168.0.1/24	down	down
GigabitEthernet1/0/0	10.1.0.1/24	up	up
GigabitEthernet1/0/1	10.2.0.1/24	up	up
GigabitEthernet1/0/2	unassigned	down	down
GigabitEthernet1/0/3	unassigned	down	down
GigabitEthernet1/0/4	unassigned	down	down
GigabitEthernet1/0/5	unassigned	down	down
GigabitEthernet1/0/6	unassigned	down	down
NULL0	unassigned	up	up (s)
Virtual-if0	unassigned	up	up (s)



The screenshot shows a PC Simulator window titled "PC1". It has tabs for "Basic Config", "Command", "MCPacket", "UdpPacket", and "Console". The "Console" tab is active, displaying the following text:

```
Welcome to use PC Simulator!

PC>ping 10.2.0.10

Ping 10.2.0.10: 32 data bytes, Press Ctrl_C to break
Request timeout!
From 10.2.0.10: bytes=32 seq=2 ttl=127 time=47 ms
From 10.2.0.10: bytes=32 seq=3 ttl=127 time=62 ms
From 10.2.0.10: bytes=32 seq=4 ttl=127 time=63 ms
From 10.2.0.10: bytes=32 seq=5 ttl=127 time=63 ms

--- 10.2.0.10 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/58/63 ms

PC>|
```

Note: Upload the **.topo** and **.cfg** files on MS Teams Assignment.